CORE PURE / PROOF BY INDUCTION

Proof by Induction - Derivatives

Mark scheme 8 questions · 58 marks

Questions

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	Total	**58**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$\frac{\mathrm{d}y}{\mathrm{d}x} = (ax^2 + 2x)e^{ax}$ $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = (a^2x^2 + 4ax + 2)e^{ax}$ $n = 2$: $a^0(a^2x^2 + 4ax + 2)e^{ax}$, so true for $n = 2$	**B1**	Second derivative found by differentiating and shown equal to the formula at $n = 2$. Not for substituting $n = 2$ into the formula alone.
	Assume true for $n = k$: $\frac{\mathrm{d}^{k+1}y}{\mathrm{d}x^{k+1}} = \frac{\mathrm{d}}{\mathrm{d}x}\left[a^{k-2}\left(a^2x^2 + 2kax + k(k-1)\right)e^{ax}\right]$	**M1**	States the result assumed for $n = k$ and takes its derivative as the $(k+1)$th derivative.
	$a^{k-2}\left[(2a^2x + 2ka)e^{ax}\right.$ $\left. + \left(a^2x^2 + 2kax + k(k-1)\right)ae^{ax}\right]$	**M1 A1**	**M1** Applies the product rule to their kth derivative: two terms, the quadratic differentiated and e^{ax} differentiated to ae^{ax}. Condone coefficient slips. **A1** Any correct unsimplified form, a^{k-2} factored out or multiplied in.
	$= a^{k-1}\left(a^2x^2 + 2(k+1)ax + k(k+1)\right)e^{ax}$ $= a^{(k+1)-2}\left(a^2x^2 + 2(k+1)ax\right.$ $\left. +(k+1)\left((k+1)-1\right)\right)e^{ax}$	**A1**	cso. Reaches $a^{k-1}(\ldots)e^{ax}$ and shows it is the formula with $n = k+1$, e.g. $k(k+1)$ written as $(k+1)((k+1)-1)$ or the target stated.
	True for $n = 2$; true for $n = k \Rightarrow$ true for $n = k + 1$ $\Rightarrow$ true for all integers $n \geq 2$	**E1**	Concludes naming true for $n = 2$, true for $n = k$ implies true for $n = k + 1$, and hence true for all integers $n \geq 2$.
	Total for Question 1	**6**	

Question	Scheme	Marks	Notes
2	$\frac{\mathrm{d}}{\mathrm{d}x}(x^3e^{-x}) = 3x^2e^{-x} - x^3e^{-x}$ $= -e^{-x}(x^3 - 3x^2)$ $n = 1$: $(-1)^1e^{-x}(x^3 - 3x^2 + 0 - 0)$, so true	**B1**	Needs both the derivative and the formula evaluated at $n = 1$, shown equal.
	Assume true for $n = k$: $\frac{\mathrm{d}^{k+1}}{\mathrm{d}x^{k+1}}(x^3e^{-x})$ $= \frac{\mathrm{d}}{\mathrm{d}x}\left[(-1)^ke^{-x}(x^3 - 3kx^2 + \ldots)\right]$	**M1**	States the assumption for $n = k$ and differentiates its expression to obtain the $(k + 1)$th derivative. Not for assuming $n = k + 1$.
	$= (-1)^k\left[-e^{-x}(x^3 - 3kx^2 + 3k(k-1)x$ $- k(k-1)(k-2)) + e^{-x}(3x^2 - 6kx + 3k(k-1))\right]$	**M1 A1**	**M1** Uses the product rule on e^{-x} times their cubic: two terms, one with the cubic's derivative. Condone sign slips. **A1** Correct, unsimplified.
	$= (-1)^{k+1}e^{-x}\big(x^3 - 3(k+1)x^2$ $+ 3(k+1)kx - (k+1)k(k-1)\big)$	**A1**	cso. Each coefficient collected to the $n = k + 1$ form with $(-1)^{k+1}$; constant $-(k + 1)k(k - 1)$ oe accepted.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k + 1$; so true for all positive integers n	**E1**	States the base case, the implication from $n = k$ to $n = k + 1$, and truth for all positive integers n.
	Total for Question 2	**6**	

Question	Scheme	Marks	Notes
3	$n=1$: $\frac{\mathrm{d}y}{\mathrm{d}x} = e^{-2x}(-2\cosh 3x + 3\sinh 3x)$ $\frac{1+(-5)}{2} = -2$, $\frac{1-(-5)}{2} = 3$, so true for $n=1$	**B1**	Correct derivative and both coefficients of the formula evaluated at $n=1$, seen to agree.
	Assume $\frac{\mathrm{d}^k y}{\mathrm{d}x^k} = e^{-2x}\left(\frac{1+(-5)^k}{2}\cosh 3x + \frac{1-(-5)^k}{2}\sinh 3x\right)$ $\frac{\mathrm{d}^{k+1} y}{\mathrm{d}x^{k+1}} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}^k y}{\mathrm{d}x^k}\right)$	**M1**	States the result assumed for $n=k$ and sets out to differentiate it to obtain the $(k+1)$th derivative.
	$\frac{\mathrm{d}^{k+1} y}{\mathrm{d}x^{k+1}} = -2e^{-2x}\left(\frac{1+(-5)^k}{2}\cosh 3x + \frac{1-(-5)^k}{2}\sinh 3x\right)$ $+ e^{-2x}\left(\frac{3(1+(-5)^k)}{2}\sinh 3x + \frac{3(1-(-5)^k)}{2}\cosh 3x\right)$	**M1 A1**	**M1** Applies the product rule to the $n=k$ expression: two terms, the second with $\cosh 3x$ and $\sinh 3x$ interchanged. Condone coefficient and sign slips. **A1** Correct, unsimplified.
	$= e^{-2x}\left(\frac{1-5(-5)^k}{2}\cosh 3x + \frac{1+5(-5)^k}{2}\sinh 3x\right)$ $= e^{-2x}\left(\frac{1+(-5)^{k+1}}{2}\cosh 3x + \frac{1-(-5)^{k+1}}{2}\sinh 3x\right)$	**A1**	cso. Both coefficients collected and reached in the $(-5)^{k+1}$ form without error; not for jumping to the printed form.
	True for $n=1$; true for $n=k \Rightarrow$ true for $n=k+1$ $\Rightarrow$ true for all $n \in \mathbb{N}$	**E1**	States all three: true for $n=1$; true for $n=k$ implies true for $n=k+1$; hence true for all $n \in \mathbb{N}$ oe.
	Total for Question 3	**6**	

Question	Scheme	Marks	Notes
4	$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x\cosh x + x^2\sinh x$ $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = x^2\cosh x + 4x\sinh x + 2\cosh x$ Compares with the formula at $n = 1$	**B1**	Both derivatives correct and matched to the formula at $n = 1$. Not for substituting $n = 1$ into the formula alone.
	Assumes $\frac{\mathrm{d}^{2k}y}{\mathrm{d}x^{2k}} = x^2\cosh x + 4kx\sinh x$ $+ 2k(2k-1)\cosh x$ $\frac{\mathrm{d}^{2k+2}y}{\mathrm{d}x^{2k+2}} = \frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\frac{\mathrm{d}^{2k}y}{\mathrm{d}x^{2k}}\right)$	**M1**	States the result for $n = k$ and takes the $(2k+2)$th derivative as its second derivative; implied by differentiating it.
	$\frac{\mathrm{d}^{2k+1}y}{\mathrm{d}x^{2k+1}} = 2x\cosh x + x^2\sinh x$ $+ 4k(\sinh x + x\cosh x) + 2k(2k-1)\sinh x$	**M1**	Differentiates their $n = k$ expression, with the product rule on at least one product and $\cosh x \to \sinh x$. Condone coefficient slips.
	$\frac{\mathrm{d}^{2k+2}y}{\mathrm{d}x^{2k+2}} = 2x\sinh x + x^2\cosh x$ $+ (4k+2)(\cosh x + x\sinh x) + 2k(2k+1)\cosh x$	**M1 A1**	**M1** Differentiates their $(2k+1)$th derivative in the same way. Condone coefficient slips. **A1** Correct, in any unsimplified form.
	$4k^2 + 6k + 2 = 2(k+1)\,(2(k+1)-1)$ $\Rightarrow \frac{\mathrm{d}^{2k+2}y}{\mathrm{d}x^{2k+2}} = x^2\cosh x + 4(k+1)x\sinh x$ $+ 2(k+1)\,(2(k+1)-1)\cosh x$	**A1**	cso. Collects to the formula with $n = k+1$, the $\cosh x$ coefficient shown as $2(k+1)(2k+1)$ or better.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k+1$; so true for all integers $n \geq 1$	**E1**	States all three: true for $n = 1$, true for $n = k$ implies true for $n = k+1$, hence true for all $n \geq 1$.
	Total for Question 4	**7**	

Question	Scheme	Marks	Notes
5	$\frac{\mathrm{d}y}{\mathrm{d}x} = 2a(ax+1)\ln(ax+1) + a(ax+1)$ $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = a^2\big(2\ln(ax+1)+3\big)$ $\frac{\mathrm{d}^3y}{\mathrm{d}x^3} = \frac{2a^3}{ax+1}$ $n=3$: $2(-1)^0\frac{0!\,a^3}{(ax+1)^1} = \frac{2a^3}{ax+1}$	**B1**	Three correct derivatives, any equivalent form, and the formula at $n=3$ shown equal to the third derivative.
	Assume $\frac{\mathrm{d}^k y}{\mathrm{d}x^k} = 2(-1)^{k-3}\frac{(k-3)!\,a^k}{(ax+1)^{k-2}}$ $\frac{\mathrm{d}^{k+1}y}{\mathrm{d}x^{k+1}} = \frac{\mathrm{d}}{\mathrm{d}x}\left(2(-1)^{k-3}(k-3)!\,a^k(ax+1)^{-(k-2)}\right)$	**M1**	States the $n=k$ assumption and attempts to differentiate it to find the $(k+1)$th derivative.
	$= 2(-1)^{k-3}(k-3)!\,a^k$ $\times\big(-(k-2)\big)(ax+1)^{-(k-1)}\times a$	**M1 A1**	**M1** Differentiates to the form $c(ax+1)^{-(k-1)}$. Condone sign slips and a missing factor a. **A1** Correct, unsimplified.
	$= 2(-1)^{k-2}\frac{(k-2)!\,a^{k+1}}{(ax+1)^{k-1}}$	**A1**	cso. Needs $(-1)^{k-2}$ from the sign and $(k-2)(k-3)! = (k-2)!$, shown or clearly implied.
	True for $n=3$; true for $n=k \Rightarrow$ true for $n=k+1$; so true for all integers $n \geq 3$	**E1**	States all three points: the base case, the implication from $n=k$ to $n=k+1$, and truth for all integers $n \geq 3$.
	Total for Question 5	**6**	

Question	Scheme	Marks	Notes
6(a)	$\frac{\mathrm{d}y}{\mathrm{d}t} = a\ln(1+at) + a$ $\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = a^2(1+at)^{-1}$, $\frac{\mathrm{d}^3y}{\mathrm{d}t^3} = \ldots$	**M1**	Differentiates repeatedly, reaching at least the second and third derivatives in the form $ka^m(1+at)^{-p}$. Condone sign and coefficient slips.
	$\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = \frac{a^2}{1+at}$, $\frac{\mathrm{d}^3y}{\mathrm{d}t^3} = -\frac{a^3}{(1+at)^2}$ $\frac{\mathrm{d}^4y}{\mathrm{d}t^4} = \frac{2a^4}{(1+at)^3}$, $\frac{\mathrm{d}^5y}{\mathrm{d}t^5} = -\frac{6a^5}{(1+at)^4}$	**A1**	All four correct, any equivalent form.
	$\frac{\mathrm{d}^ny}{\mathrm{d}t^n} = \frac{(-1)^n(n-2)!\,a^n}{(1+at)^{n-1}}$	**A1**	oe, such as $(-1)^{n-2}$ or $(1+at)^{1-n}$ in the numerator. Must be in terms of n, a and t.
		[3]	
6(b)	$n=2$: $\frac{(-1)^2\,0!\,a^2}{(1+at)^1} = \frac{a^2}{1+at} = f''(t)$	**B1**	Shows the formula with $n=2$ gives $\frac{a^2}{1+at}$ and that this matches $f''(t)$.
	Assume $f^{(k)}(t) = \frac{(-1)^k(k-2)!\,a^k}{(1+at)^{k-1}}$ $f^{(k+1)}(t)$ $= \frac{\mathrm{d}}{\mathrm{d}t}\left[(-1)^k(k-2)!\,a^k(1+at)^{-(k-1)}\right]$	**M1**	States the formula assumed for $n=k$ and sets out to differentiate it to obtain $f^{(k+1)}(t)$.
	$= (-1)^k(k-2)!\,a^k$ $\times(-(k-1))\,a(1+at)^{-k}$	**M1 A1**	**M1** Differentiates the assumed result, reducing the power of $(1+at)$ by 1 with a multiplying factor. Condone sign and coefficient slips. **A1** Correct derivative, unsimplified, including the factor a.
	$= \frac{(-1)^{k+1}(k-1)!\,a^{k+1}}{(1+at)^k}$	**A1**	cso. Combines $(k-1)(k-2)!$ as $(k-1)!$ and the signs as $(-1)^{k+1}$, giving the formula with $n=k+1$.
	True for $n=2$; true for $n=k$ $\Rightarrow$ true for $n=k+1$; so true for all integers $n \geq 2$	**E1**	Concludes, naming true for $n=2$, true for $n=k$ implies true for $n=k+1$, and hence true for all integers $n \geq 2$.
		[6]	
6(c)	$f^{(7)}\left(\frac{5}{3}\right) = \frac{(-1)^7\,5!\times3^7}{\left(1+3\times\frac{5}{3}\right)^6} = \ldots$	**M1**	Substitutes $n=7$, $a=3$ and $t=\frac{5}{3}$ into their formula. Not for the sixth derivative.
	$-\frac{45}{8}$	**A1**	oe, or awrt -5.63.
		[2]	
	Total for Question 6	**11**	

Question	Scheme	Marks	Notes
7(a)	$f'(x) = -\frac{1}{2}(1+x)^{-\frac{3}{2}}$ $\frac{(-1)^1\,2!}{2^2\,1!\,(1+x)^{\frac{3}{2}}} = -\frac{1}{2}(1+x)^{-\frac{3}{2}}$	**B1**	Both the derivative and the formula at $n=1$, shown equal.
	Assume $f^{(k)}(x) = \frac{(-1)^k(2k)!}{2^{2k}k!(1+x)^{k+\frac{1}{2}}}$ $f^{(k+1)}(x) = \frac{(-1)^k(2k)!}{2^{2k}k!} \times \frac{\mathrm{d}}{\mathrm{d}x}(1+x)^{-k-\frac{1}{2}}$	**M1**	States the formula assumed for $n=k$ and sets out to differentiate it for $f^{(k+1)}(x)$. Not for assuming the $n=k+1$ result.
	$= \frac{(-1)^k(2k)!}{2^{2k}k!}\left(-k-\frac{1}{2}\right)$ $\times (1+x)^{-k-\frac{3}{2}}$	**M1 A1**	**M1** Differentiates their $f^{(k)}(x)$ to the form $c(1+x)^{-k-\frac{3}{2}}$. Condone sign and coefficient slips. **A1** Any unsimplified form.
	$= \frac{(-1)^{k+1}(2k+2)(2k+1)(2k)!}{2^{2k+2}(k+1)k!(1+x)^{k+\frac{3}{2}}}$ $= \frac{(-1)^{k+1}(2k+2)!}{2^{2k+2}(k+1)!(1+x)^{k+\frac{3}{2}}}$	**A1**	cso: the $n=k+1$ formula reached with $(2k+2)!$ and $(k+1)!$ formed explicitly; any order of steps. Jumping to the result scores A0.
	True for $n=1$; true for $n=k \Rightarrow$ true for $n=k+1$ $\Rightarrow$ true for all $n \geq 1$	**E1**	States all three points; condone equivalent wording such as "for all positive integers n".
		[6]	
7(b)	$f(0)=1, \quad f^{(n)}(0) = \frac{(-1)^n(2n)!}{2^{2n}n!}$	**B1**	Both, with $f^{(n)}(0)$ from part (a).
	$\frac{x^n}{n!}f^{(n)}(0) = \frac{x^n}{n!} \times \frac{(-1)^n(2n)!}{2^{2n}n!}$	**M1**	Substitutes their $f^{(n)}(0)$ into $\frac{x^n}{n!}f^{(n)}(0)$, the Maclaurin general term quoted or implied.
	$= (-1)^n \frac{(2n)!}{2^{2n}(n!)^2}x^n$ $(1+x)^{-\frac{1}{2}} = 1 - \frac{x}{2} + \frac{3x^2}{8} - \cdots$	**A1 AG**	Correct substituted general term, combining the two $n!$ to $(n!)^2$, and the printed series stated without error.
		[3]	
	Total for Question 7	**9**	

Question	Scheme	Marks	Notes
8	$\frac{\mathrm{d}^2}{\mathrm{d}x^2}(x^2f) = 2f + 4xf' + x^2f''$ $= 2f + 4xf' - x^2f$ $= -(x^2f - 4xf' - 2f)$	**B1**	Differentiates twice, uses $f'' = -f$ and shows agreement with the RHS at $n = 1$. Not for substituting $n = 1$ into the RHS alone.
	Assume $\frac{\mathrm{d}^{2k}}{\mathrm{d}x^{2k}}(x^2f)$ $= (-1)^k(x^2f - 4kxf' - 2k(2k-1)f)$ $\frac{\mathrm{d}^{2k+2}}{\mathrm{d}x^{2k+2}}(x^2f)$ $= (-1)^k\left[\frac{\mathrm{d}^2}{\mathrm{d}x^2}(x^2f) - 4k\frac{\mathrm{d}^2}{\mathrm{d}x^2}(xf')\right.$ $\left.-2k(2k-1)f''\right]$	**M1**	Assumes the result for $n = k$ and writes the $(2k+2)$th derivative as the second derivative of the assumed expression.
	$\frac{\mathrm{d}^2}{\mathrm{d}x^2}(x^2f) = 2f + 4xf' + x^2f''$ $\frac{\mathrm{d}^2}{\mathrm{d}x^2}(xf') = 2f'' + xf'''$	**M1**	Finds the second derivatives of x^2f (or quotes the base case) and xf' by the product rule; whole-expression differentiation also scores. Condone slips.
	$f'' = -f,\ f''' = -f' \Rightarrow \frac{\mathrm{d}^{2k+2}}{\mathrm{d}x^{2k+2}}(x^2f) = \ldots$	**M1**	Replaces f'' by $-f$ and f''' by $-f'$ throughout and collects terms in x^2f, xf' and f. Condone sign and coefficient slips.
	$= (-1)^k(-x^2f + 4(k+1)xf'$ $+ (2 + 8k + 2k(2k-1))f)$	**A1**	Any correct unsimplified form in f and f' only.
	$= (-1)^{k+1}(x^2f - 4(k+1)xf'$ $- 2(k+1)(2k+1)f)$	**A1**	cso, with the coefficient of f seen as $2(k+1)(2k+1)$ or $2(k+1)(2(k+1)-1)$.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k + 1$; so true for all positive integers n	**E1**	States true for $n = 1$, true for $n = k$ implies true for $n = k + 1$, hence true for all positive integers n.
	Total for Question 8	**7**	